- Polynesia includes Hawai, New Zealand, Tonga island, Easter Island
- Places: Great Dividing Range, Great Barrier Reef, Mt Wilhelm, Southern Alps
- Antarctica
- Antarctic treaty System, 1961 held that Antarctica can be used only for peaceful experiments.
- The base stations of India in Antarctica are **Dakshin Gangotri, Maithri, and Bharati.**
- **India (7:38 PM)**
- We have both land and maritime neighbours.
- Land Neighbours are Pakistan, Afghanistan, China, Nepal, Bhutan, Bangladesh, and Myanmar.
- Afghanistan shares a border with India along **Wakhan Corridor**.
- Countries arranged in order of length of the border with India: Bangladesh(4000 km)> China(3500 km)>Pakistan(3300 km)>Myanmar(1700 km)>Nepal(1650 km)>Bhutan(700 km)>Afghanistan(100 km)
- Countries with maritime boundaries: Bangladesh, Pakistan, Myanmar, Sri Lanka, Maldives, Indonesia, Thailand
- Union territories of India: Andaman and Nicobar islands, The Government of NCT of Delhi, Ladakh, Puducherry, Lakshadweep, Jammu and Kashmir, Chandigarh, and Dadra and Nagar Haveli and Daman and Diu.
- The highest peak in India is **K2** and the Second highest peak is **Kanchenjunga**
- The lowest point in India is in **Kuttanad** (Kerala)

The topic for the next class: Physical geography of India

Geography Class 36

A brief discussion of the Previous Class (5:11 PM)

Mapping(5:25 PM)

- Sikkim and Meghalaya share a border with only one state.
- Sikkim shares a border with West Bengal only. Meghalaya Shares a border with Assam only.
- Telangana doesn't share a border with Odisha.
- Madhya Pradesh doesn't share a border with Jharkhand.
- Yamuna flows through Himachal Pradesh as well.

Physiography of India (7:53 PM)

Subcontinent

- Subcontinent is the large landmass i.e. part of the continent but is considered separate from the rest of the continent.
- For example, the Indian subcontinent, Alaskan Subcontinent, etc.
- Reasons for considering India as a subcontinent.
- (a) Geographical reasons
- Geographically, the Indian subcontinent has well-defined physical boundaries, such as the Himalayas and the Hindukush in the north, Purvanchal in the east, and the Indian Ocean in the South.
- (b) Geological reasons
- The countries of the Indian subcontinent share common geological history, such as the formation of the Himalayas, northern plains, etc.
- (c) Common climatic conditions
- All these countries experience monsoon types of climate.

- (d) Political reasons
- They share a common political identity i.e. South Asia.
- (e) Historical and Cultural reasons
- They share a common history, ruled by large empires such as Mauryan, Mughal, etc.
- They also share a common culture of language, festivals, etc.

Mountains of India (6:05 PM)

Himalayas

- **Origin of the Himalayas**
- (Discussed in the Plate tectonic Class. Please refer to the notes of those classes.)
- Different stages in the formation of the Himalayas
- The Himalayas were formed in three different phases,
- (a) First Stage between 120-70 million years ago
- During this stage, great Himalayas were formed.
- (b) Second stage between 30-25 million years ago.
- This resulted in the formation of the middle Himalayas.
- (c) Third stage between 20-2 million years ago
- This resulted in the formation of the Shivaliks.
- **Syntaxial Bending**
- It refers to a deep knee bent, with the strike of the ridges suddenly turning at right angles.
- It is due to the maximum push offered at both ends of the Indian Peninsula during its northward drift.
- In the northwest direction, the Aravalis and in the northeast the Assam ranges acted as two extended arms, while the central area sagged giving an arcuate shape.
- **Evidence to prove that the Himalayas are still rising**
- (a) Frequent earthquakes
- (b) Recent adjustment in the height of Himalayan peaks.
- (c) Frequent disturbance in the topographies.
- (d) The rivers of the Himalayas, are in the youthful stage over a long time.
- (e) Drying up of lakes of Tibet, which shows that the Tibetan plateau is still rising.
- (Kindly refer to the diagram drawn by the faculty in the class for understanding the Fracture Zones of the Himalayas)

Regional Division of Himalayas (7:17 PM)

- The Himalayas are divided by rivers. This classification is called regional division. The main classifications are:
- (a) Kashmir Himalayas
- Between the river Sutlej and Indus
- (b) Kumaon Himalayas
- Between the river Sutlej and Kali
- (c) Nepal Himalayas
- Between the river Kali and Teesta
- (d) Assam Himalayas
- Between the river Teesta and Brahmaputra.
-

Western Himalayas	Eastern Himalayas
It includes Punjab and Kumaon Himalayas.	It includes Sikkim and Arunachal Himalayas.

It is between River Indus and River Kali.	It is between River Teesta and the Brahmaputra.
These are wider.	These are narrower.
Middle Himalayas and Shivaliks are separate.	These have merged in the eastern Himalayas.
These are taller.	These are comparatively shorter.
These rise gradually.	These rise abruptly.
The temperature of the western Himalayas is comparatively higher.	The temperature of the eastern Himalayas is comparatively higher.
Coniferous type of vegetation.	Evergreen and deciduous types of vegetation.
Lower biodiversity.	Higher biodiversity
Lower precipitation.	Higher precipitation.
Snowline at the lower altitude	Snowline at the higher altitude.

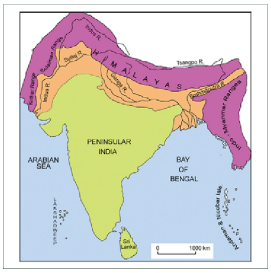
Longitudinal Division (7:36 PM)

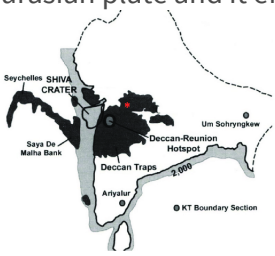
- (a) Trans Himalayas
- Karakoram range, Zaskar range, Ladakh range and Kailash range.
- (b) Great Himalayas
- It extends from Nanga Parbat to Namcha Barwa.
- These are also known as Himadri.
- (c) Middle Himalayas/ Lesser Himalayas/ Himachal
- These are discontinuous ranges.
- Pir Panjal range, Dhauladhar range, Mahabharat range, etc.
- the average height is 4000 m.
- (d) Shivaliks
- The average height is just 1000m.
- Dafla, Mirim Abhor, and Mishmi are parts of the Middle Himalayas and Shivaliks.
- (e) Purvachal
- These are hills of the northeast.
- It includes Mizo hills, Tripura Hills, Naga Hills, etc.

The Topics for the Next Class: Mountains of India (Continued)
Geography Class 37

Geography Class 37 [17:11:00]
A brief overview of the previous class i.e. Himalayas and Q&A session
The Peninsular Block [17:43:00]

- The peninsular block is a great complex of Ancient rocks which has existed as a single rigid block for Millions of years
- It extends from Gujarat in the west to Meghalaya in the east and from Plains in the north to the southern tip of India in the south
- [* Why did Meghalaya get separated?- A gap was formed named as "**Malda gap**" or "Garo-Rajmahal gap"]

- 

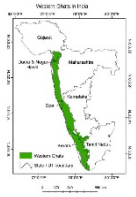
- This region was subjected to a few episodes of Tectonic activity
- 1) Submergence of the Western part of Western Ghats
- 2) Formation of Rift Valleys such as Narmada-Tapi etc
- 3) Formation of Deccan traps when the Indian plate was moving towards the Eurasian plate and it encountered the Reunion Hotspots
- 

Mountain ranges in Peninsular Block [18:02:00]

- Arvallis- Old fold mountains of earlier times
- Vindhyas and Satpuras- Block mountains
- From north to south- Vindhya> Satpura> Narmada> Tapi
- Western Ghats
- Eastern Ghats

Western Ghat and Eastern Ghat comparison [18:05:00]

-

	Western Ghats	Eastern Ghats
Extent	Gujarat (south of Tapi) to kanyakumari	South of Mahanadi to Nilgiri
States	Gujarat, Maharashtra, Goa, Karnataka, Kerala, Tamilnadu 	Orissa, Andhra Pradesh, Telangana, Tamilnadu, Karnataka

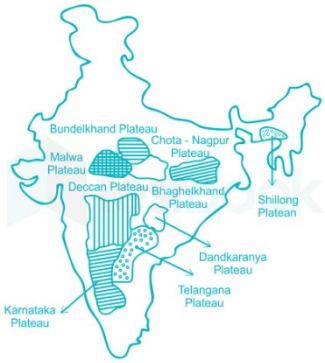
Continuity	Continuous	Discontinuous [* Due to the rivers coming from the west which has eroded the eastern Ghats]
Slope	These are taller	These are a bit shorter
	Narrow	Wide
Topography	Highest peak- Anaimudi Overall height increases towards the south	Highest peak- Mahendragiri Some sources state that Zindgada's peak Height increases towards the North
Closeness with coast	Closer to the coast	A Bit far from the coast
Physiography	More river originates from western Ghat	Less river originates from eastern Ghat
Precipitation	Higher precipitation	Lower precipitation
Vegetation	Vegetation in Western ghat is an evergreen type	Vegetation- Deciduous
Biodiversity	Higher Biodiversity in Western Ghat	Lower biodiversity in Eastern Ghat

- Ghat means Pass [Palghat, Bhorghat] or highly elevated regions
- **Parts of Western Ghats**

- In Maharashtra and Karnataka- Called Sahayadri
- In Karnataka- Baba Budan Hills. It is the origin of coffee
- Nilgiri- Because of Neelkurinji Flowers
- Anaimalai Hills
- Cardamom Hills
-

- **Parts of Eastern Ghats**
- Mahendragiri
- Nallamala hills
- Palkonda Hills- Tirupati is part of this
- Javadi & Shevroy hills
-

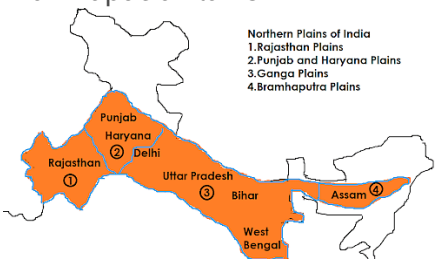
Plateaus of Peninsular Block [18:25:00]

- Malwa plateau
- **Bundelkhand Plateau**
- 

- **Chotanagpur Plateau-** Entire Jharkhand state
- **Meghalaya Plateau**- Garo, Khasi, and Jaintia hills are part of this plateau
- **Deccan Plateau**- Between Western Ghat and Eastern Ghat- Maharashtra, Karnataka, and Northern part of Telangana are covered in Deccan traps
- **Rayalseema Plateau**- Region within Andhra Pradesh. [* It is called Rayalseema because it was ruled by the Clan of Rayas]
- **Telangana plateau**
- [Krishna river marks the boundary between Rayalseema and Telangana plateau]

- **Karnataka plateau**- Towards western ghat, it is a hilly region called **Malnad** and on the other side it is a plain region called Rolling plain/ **Maidan**
- **Dandakarnya plains**

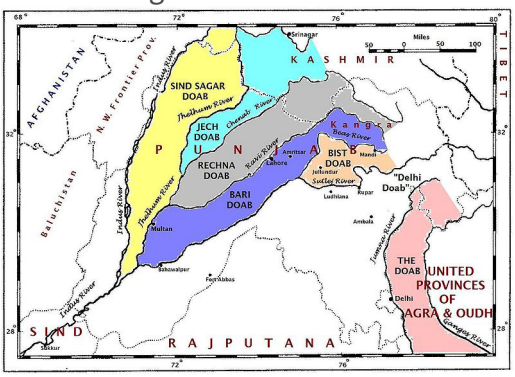
Plains of India [18:36:00]

- Flat regions
- Two types- Northern Plains and Coastal Plains
- **Northern Plains**
- **How it originated?**
- The Northern Plains are formed by the deposition of sediments brought by the Himalayan rivers into the basin between the Himalayas and Peninsular blocks.
- It extends for a total of 3200 KMs of which plains of India are for 2400 KMs
- This is the largest alluvial plain in the world
- It is divided into the Rajasthan plain, Punjab& Haryana Plains, Ganga plains, and Brahmaputra Plains
- 

Rajasthan Plains [19:10:00]

- It is a riverine plain
- It is formed due to sediments brought by the rivers
- They may have formed due to the drying up of the rivers which once existed in this region
- The western part of Rajasthan is a proper desert called "Marusthali"
- The eastern part of Rajasthan plains is called Rajasthan Bagar which is semi-arid due to the presence of river Luni

Punjab & Haryana Plains [19:20:00]

- These are the plains of the Indus
- Doab is a region between two rivers
- 

- BIST doab- between Beas and Sutlej rivers. It is only in India.
- BARI doab- Between Beas and Ravi rivers
- RECHNA doab- Between Ravi and Chenab
- CHAJ doab- Between Chenab and Jhelum
- Sind Sagar Doab- Between Sind and Chenab

Ganga plains [19:26:00]

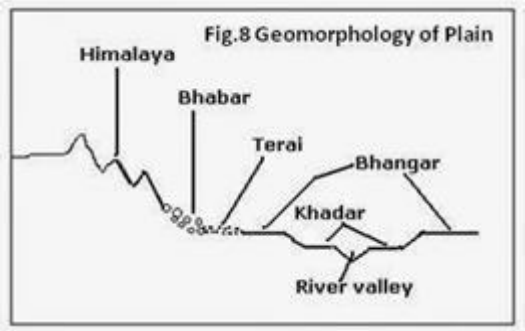
- It covers UP, Bihar, and West Bengal

- It extends from Delhi to Kolkata
- **Parts of the Ganga plains**
- Doab- Ganga- Yamuna Doab
- Rohilkhand plains
- Awadh plains
- Lower Ganga plains

Brahmaputra Plains [19:31:00]

- It is also called the Assam plains
- Extent- From Sadiya in the east to Dhubri in west
- It is surrounded by Hills

Types of Northern Plains [19:35:00]

- **Bhabhar-**
- It is a region of Porous gravel and boulders deposited by rivers entering the plains from the Mountains
- In this region the smaller rivers disappear and start flowing underground
- **Terai-**
- It is a marshy region after Bhabhar where rivers start to re-appear. It is very fertile and widely used for the agriculture of rice and sugarcane
- They are waterlogged regions and are prone to **water-borne diseases**/ **Mosquito-borne diseases**
- Northern parts of UP, Bihar border, Nepal, Muzaffarnagar area.
- 

- **Khadar**
- It is the alluvial plain just next to River valley.
- It is made up of fresh alluvial soil which gets replenished every year
- It is more fertile
- **Bangar**
- It is the older alluvial plain located away from the river valley
- It is less fertile than Khadar
- It is more settled.

Duars formation [19:50:00]

- These are unconsolidated sediments deposited by rivers from mountains along the foothills of the North-east region i.e. West Bengal and Assam
- They are best suited for tea cultivation

Coastal Plains [19:53:00]

- They are formed by the sediments deposited by the rivers coming from the peninsular region
- There are two coastal plains- The eastern coastal plains and The western coastal Plains
-

Western coastal plain	**Eastern Coastal Plain**
From Kutch to Kanyakumari	From Sundarbans to Kanyakumari
Narrow [* The length covered by rivers is less thus fewer sediments]	Wider [* Rivers flow for long distances and carry large sediments]
The maximum extent is 65 Km	Maximum extent is 150-200 Km
Dominated by Estuaries	Dominated by Deltas
The western coast is called a submerged coast. However, the Malabar coast is emergent one	The entire Eastern coast is emergent
The erosion process is more active	The deposition process is active- So more beaches, sandbars
The western coast is a broken coast or an Indented coast	The eastern coast is smooth. The coastline is smooth
Natural harbours and ports because of the broken coast and erosion.	Few of them are Natural harbours but most are artificial/ man-made port
Wider continental shelves around the western coast	Narrow continental shelves around the eastern coast
Kathiawad coast, Konkan coast, Malabar coast 	Utkal plain, Northern circar, Coromanadal coast

The topic for the next class:- Drainage systems and Islands

Geography Class 38

DISCUSSION ON THE UPSC 2023 PRELIMS PAPER (5:16 PM)

DISCUSSION ON THE PREVIOUS CLASS (5:44 PM)
DRAINAGE SYSTEM OF INDIA (5:49 PM)

- The drainage system of India is mainly divided into Himalayan rivers and Peninsular rivers.
- **Evolution of Himalayan Drainage System(5:55 PM)**
- Widely accepted theory regarding the evolution of Himalayan drainage proposes the existence of a single river all along the entire stretch of Himalayas to the south of Shivaliks flowing from East to West called Indobrahma or Shivalik River.
- It drained into the Gulf of Sindh.
- The earlier drainage system was disturbed by three main events:
- (a) Higher upliftment of the western Himalayas.
- (b) Rising of Potwar plateau and Delhi ridge.
- (c) Downthursting of Malda gap or Garo-Rajmahal gap.
- This divided the entire river into Indus and its tributaries, Ganga and its tributaries, and the Brahmaputra and its tributaries.
- **Peninsular Drainage System (6:08 PM)**
- The peninsular drainage system is influenced by three major events:
- (a) Submergence of the western side of the western ghats which disturbed the earlier symmetrical pattern of peninsular drainage and shortened the west-flowing rivers.
- (b) Upliftment of the Himalayas which caused the northern part of the Vindhyas to tilt more towards the north and also the deepening of the rift valley of Narmada and Tapi.
- (c) Slight tilting of the peninsular block from northwest to southeast direction, causing the slope towards the Bay of Bengal drainage.
- **Difference between Himalayan and Peninsular Drainage (6:26 PM)**
-

Himalayan Drainage	**Peninsular Drainage**
The source of the drainage is glaciers.	The source of the drainage is rainfall.
The Himalayan drainage is perennial.	The Peninsular Rivers are seasonal.
The Himalayan rivers are dominated by the youth stage.	The Peninsular rivers are dominated by mature and old stages.
They form V-shaped valleys.	They formed U-shaped valleys.
They have high meandering. (High meandering due to high sediments and flowing through plain regions.)	These have low meandering. (Low meandering due to lower sediments and flowing through plateau region.)

These have larger river basins and a catchment area.	These have smaller river basins and a catchment area.
The deltas of the Himalayan rivers are very large.	The deltas present are small. Some rivers just have estuaries.
Himalayan rivers are more navigable in the plains.	These are less navigable.
The Himalayan drainage has many antecedent rivers.	In Peninsular drainage, there are only consequent rivers.
The drainage pattern is mainly dendritic.	The drainage pattern is mainly rectangular or trellis.

- Examples of antecedent rivers are Indus, Brahmaputra, Sutlej, Ghagra, Gandak, and Kosi. Ganga and Yamuna are the consequent rivers.

DRAINAGE PATTERNS (7:06 PM)

- The flow characteristic of a river is called a drainage pattern.
- (a) Dendritic Pattern
- It appears like the branches of a tree or a leaf.
- This is characterized by the lack of structural control.
- This is found in the Himalayan rivers. For example, the Ganga drainage pattern.
- (b) Rectangular Pattern
- The river takes sharp turns.
- It is found in the plateau regions with high structural control.
- This is found in the case of the peninsular rivers. For example, the River Mahanadi, Godavari, Cauvery, etc. follows this drainage pattern.
- (c) Trellis pattern
- The main river flows in the rift valley and tributaries join at the right angle.
- This is possible when the river flows in the rift valley. For example, the Narmada and Tapi Rivers follow this drainage pattern.
- (d) Radial Pattern
- In this, the different rivers originate from the raised platform or mountain and flow in various directions.
- For example, from Amarkantak we have the Narmada, Son, and Hasdeo rivers originate.
- (e) Parallel Drainage
- The rivers flow parallel to each other and drain directly into the sea.
- For example, West flowing rivers of the western ghats.

MAPPING OF THE RIVERS (7:28 PM)

- (a) Indus River System
- Indus flows through three countries namely China, Pakistan, and India.
- It originates near Mansarovar Lake.
- The right bank tributaries are Shyok, Gilgit, and Kabul.